

COLIDE

Current state, GitHub map, and manuscript handoff — a reading pack

Prepared 15 August 2026 for Ibteshamul Haque (and for sharing with Cheranrach Mahandren as manuscript lead). This PDF restates what is true on disk and on GitHub. It does not invent new experiments. If a number here disagrees with `docs/RESULTS_INDEX.md` or a gate JSON, the JSON wins.

1. One-page verdict

You can move to manuscript writing. Core data validity and local CUDA correctness are closed with artifacts. Cheran can lead the paper from the handoff pack. The project is **not camera-ready** until the manuscript PDF/venue draft is polished from those sources — that is writing work, not a new science campaign.

Layer	Status
Principal BoT-IoT detection	Ready — 0.9780 ± 0.0033 (n=5). Champion md5 80a90f7... unchanged.
ToN-IoT (secondary)	Ready — leakage-safe CNN 0.8075 / RF 0.9626. Clean 0.9526 path INVALID.
CUDA Block-3 source + local parity	Closed locally — GPU vs PyTorch full-seq max abs $\sim 3.43\text{e-}6$.
Local sanitizers (RTX 3050)	Closed — racecheck / synccheck / initcheck / memcheck, 0 errors.
Server B3 latency (V100S / A100)	Option B — historical pre_fix only. No post-fix DICC rebench.
Claim hygiene + GitHub navigation	Polished — stale files kept and labeled, not deleted.
Manuscript draft (Markdown)	Synced to locked numbers; figures regenerated from artifacts.
Camera-ready PDF / venue format	Open — Cheran leads writing from the pack.

GitHub: <https://github.com/titoatwork/COLIDE> · branch `master` · tip at polish `f669d4f` (public, Academic Research license).

2. Locked numbers (use only these as principal)

2.1 BoT-IoT

Item	Value
Principal test macro-F1	0.9780 ± 0.0033 (sealed multi-seed, n=5)
Champion file	model/best_model_botiot_twostage.pth
Champion MD5	80a90f7cc210276300eaa90173a5a385
Historical 0.9790	Development / legacy only — not the principal result

2.2 ToN-IoT (secondary)

Item	Value
Protocol	toniot_leakage_safe_v1
CNN test macro-F1	0.8075
RF test macro-F1	0.9626
Split	Seed 42, stratified random 60/20/20 — not official temporal/host split

Item	Value
Weak class	mitm — F1 $\approx$ 0.111 (precision $\sim$ 0.059, recall $\sim$ 0.909)
Invalid “clean” path	CNN 0.9526 / RF 0.9851 / +15.4% — INVALID (label leakage)

2.3 CUDA / systems wording

Item	How to write it
Local B3 parity	Closed. Gate JSON valid=true, kernel_status=post_fix.
DICC B3 microseconds	Historical pre_fix wall-clock only (Option B). Not a post-fix server campaign.
Full custom CUDA vs full V3	Forbidden. Option A = Blocks 1–4 only.
Framework tables	Table A (CUDA blocks, absolute) vs Table B (full model, absolute). No cross-table speedups.
$\sim$ 25,899 flows/s	Bulk batched throughput only — not streaming arrivals.
Energy	Exploratory board-power language.
LLM 16.60 μ s p99	Alert dispatch overhead only — not free-form XAI quality.
Native TensorRT equivalence	Not validated — do not claim.

3. How to read the public GitHub

The repo is public academic research. Stale or internal files were **not removed**. They are labeled so a visitor is not misled. If two documents disagree, **docs/RESULTS_INDEX.md** and **docs/CLAIM_MAP_PREWRITE.md** win.

3.1 Start here

If you are...	Open first
Cheran / writing the paper	docs/CHERAN_MANUSCRIPT_HANDOFF.md
Checking what may be claimed	docs/CLAIM_MAP_PREWRITE.md and docs/RESULTS_INDEX.md
Reading results only	README abstract + Results Summary; JSON under benchmarks/results/
Reviewing remediation	COLIDE_Remediation_Update_Review.md
Lost in the docs folder	docs/README.md (current / historical / correspondence)
Wondering which .pth is live	model/README.md

3.2 Folder map

Path	What it is
docs/	All writing. Index in docs/README.md.
docs/manuscript/	Draft manuscript Markdown + figures.
scripts/	Training, benches, parity gates, stale-claim guard.
inference/kernels/	Option A CUDA sources (+ laptop binaries).
model/	Weights: champion, corrected ToN, historical, INVALID clean.
benchmarks/results/	Claim-eligible JSON/gates (many raw benches gitignored).

Path	What it is
tests/	pytest — 28 passing at last check.
dicc_scripts/	UM DICC campaign helpers.
HANDOFF.md, DAILY_LOG.md, AGENTS.md, CLAUDE.md	Internal / historical session notes — not paper authority.

3.3 What was kept but is not current

These remain on GitHub for audit. Each has a short HISTORICAL or INTERNAL banner:

- Old professor emails (docs/EMAIL_*) — correspondence, not evidence.
- docs/PROF_POR_*, STATUS_REPORT_DRAFT, execution_plan/, audit/ — older packs.
- Root HANDOFF.md still says “pre-manuscript fully closed” in the body — that is **Aug 12 historical**; the banner says not to treat it as live.
- Long remediation checklist at repo root — some boxes lag later gate JSONs.
- model/best_model_toniot_clean.pth — INVALID (DATA-TON-001); labeled in model/README.md.

Not uploaded on purpose: large untracked experiment trees (ablation/HPO/KD), raw datasets, .venv, zip packs, most logs.

4. Sharing with Cheran (manuscript lead)

Cheran leads structure and prose. You support numbers, CUDA/DICC wording, and fact-checks. Prof. Por is supervisor only — keep him updated, do not make the pack internship-centric.

4.1 Two ways to send

GitHub (full tree): <https://github.com/titoatwork/COLIDE> — pull **master**. Tell him to open **docs/CHERAN_MANUSCRIPT_HANDOFF.md** first.

Zip (~2.1 MB): C:\Users\Ibteshamul Haque\Downloads\COLIDE_Cheran_manuscript_pack_20260815.zip (also in the repo root). Draft, figures, claim docs, key JSON — not huge training trees.

4.2 Email draft

Ready text: **docs/EMAIL_CHERAN_MATERIALS_HANDOFF.md**. To: Cheranrach Mahandren. CC: Prof. Por (recommended). Subject: COLIDE paper materials for manuscript preparation (handoff pack).

4.3 What he should use while writing

Asset	Path
Start-here map	docs/CHERAN_MANUSCRIPT_HANDOFF.md
Draft to edit	docs/manuscript/CAD_CBA_v1_MANUSCRIPT.md
Figures	docs/manuscript/figures/ (incl. ToN per-class)
Tables from JSON	docs/manuscript/TABLES_FROM_ARTIFACTS.md
OK / FORBIDDEN claims	docs/CLAIM_MAP_PREWRITE.md

Asset	Path
B3 server decision	docs/B3_SERVER_LATENCY_DECISION.md (Option B)
Corrected ToN JSON	benchmarks/results/toniot_corrected/summary.json
B3 parity gate	benchmarks/results/block3_parity_gate.json

5. Hard rules for the paper

- 1. Principal accuracy is sealed **0.9780 ± 0.0033**, multi-objective — not pure-F1 SOTA versus protocol LightGBM.
- 2. Never revive ToN clean **0.9526 / 0.9851 / +15.4%** as valid.
- 3. Never ratio incomplete Custom CUDA (Blocks 1–4) against full-model eager / compile / TensorRT as end-to-end speedup.
- 4. Never present DICC Block-3 timings as post-fix unless a new SUCCESS rebench exists (current decision: Option B).
- 5. Local B3 parity is not a server latency claim.
- 6. Architecture is a latent pseudo-sequence — not raw packet chronology.
- 7. Title/abstract: detection + systems/CUDA Option A + scoped dispatch — not production LLM explainability.

6. What is still open (honest)

Item	Need
Venue manuscript PDF / Word / LaTeX	Cheran leads from synced Markdown.
Post-fix DICC B3 latency	Only if you later reverse Option B and rebench.
Native TensorRT numerical gate	Skipped — no engine. Do not claim equivalence.
Formal multi-hundred determinism campaign	Optional. Do not claim full determinism under all loads.
Rewriting every historical email	Not required; banners already mark them internal.

7. Suggested next 48 hours

- 1. Read this PDF once through, then skim CHERAN_MANUSCRIPT_HANDOFF.md on GitHub.
- 2. Send Cheran the email + GitHub link and/or the zip from Downloads.
- 3. Agree venue and template (Word vs LaTeX) with him.
- 4. You stay on fact-check / CUDA wording; he owns narrative.
- 5. Do not retrain the champion or reopen ToN leakage unless a new review finds a real error.

Authority chain: gate JSON → RESULTS_INDEX / CLAIM_MAP → manuscript Markdown → this reading pack. Tests at last polish: 28 passed. Stale-claim guard: OK on configured active surfaces. License: Academic Research Only (LICENSE file).